Tutorial 1

Topics on Counting and Probability

HE, Yucheng

The School of Data Science(SDS)
The Chinese University of Hong Kong, Shenzhen

STA2001H, 2026

Outline

1. Introduction
2. About HW1
3. Practice

About the Teaching Teams

Below is the teaching team's contact information. Please do not hesitate to reach out to them if you need assistance.

Ajay Jasra(Prof.)	ajayjasra@cuhk.edu.cn	
	511B, Dao Yuan	14:00, Thursday
CHEN, Zhengwei(TA)	121090079@link.cuhk.edu.cn	
CHEN, Zhiyu(TA)	223040018@link.cuhk.edu.cn	
HE, Yucheng(USTF)	123030023@link.cuhk.edu.cn	
ZHAO, Runkai(USTF)	123090863@link.cuhk.edu.cn	

Remark: Please make appointment in advance!

The WeChat Group

群聊：25Spring Sta2001H学生群

该二维码7天内(1月28日前)有效，重新进入将更新

Question 5

For events A,B F, under what conditions does the equation

$$\mathbb{P}(A) = \mathbb{P}(A|B) + \mathbb{P}(A|B^c)$$

hold ?

Counting(Textbook, CH1 EX.7)

Two chess players, A and B, are going to play 7 games. Each game has three possible outcomes: a win for A (which is a loss for B), a draw (tie), and a loss for A (which is a win for B). A win is worth 1 point, a draw is worth 0.5 points, and a loss is worth 0 points.

(a) How many possible outcomes for the individual games are there, such that overall player A ends up with 3 wins, 2 draws, and 2 losses?

(b) How many possible outcomes for the individual games are there, such that A ends up with 4 points and B ends up with 3 points?

(c) Now assume that they are playing a best-of-7 match, where the match will end when either player has 4 points or when 7 games have been played, whichever is first. For example, if after 6 games the score is 4 to 2 in favor of A, then A wins the match and they don't play a 7th game. How many possible outcomes for the individual games are there, such that the match lasts for 7 games and A wins by a score of 4 to 3?

Story Proof(Textbook, CH1 EX.20)

(a) Show using a story proof that

$$\sum_{i=k}^{n} \binom{i}{k} = \binom{n+1}{k+1}$$

where n and k are positive integers with n k. This is called the hockey stick identity. Hint: Imagine arranging a group of people by age, and then think about the oldest person in a chosen subgroup.

(b) Suppose that a large pack of Haribo gummi bears can have anywhere between 30 and 50 gummi bears. There are 5 delicious flavors: pineapple (clear), raspberry (red), orange (orange), strawberry (green, mysteriously), and lemon (yellow). There are 0 non-delicious flavors. How many possibilities are there for the composition of such a pack of gummi bears? You can leave your answer in terms of a couple binomial coefficients, but not a sum of lots of binomial coefficients.

Inclusion-exclusion(Textbook, CH1 EX.52)

A certain class has 20 students, and meets on Mondays and Wednesdays in a classroom with exactly 20 seats. In a certain week, everyone in the class attends both days. On both days, the students choose their seats completely randomly (with one student per seat). Find the probability that no one sits in the same seat on both days of that week.

Conditional Probability(Textbook, CH2 EX.25)

A crime is committed by one of two suspects, A and B. Initially, there is equal evidence against both of them. In further investigation at the crime scene, it is found that the guilty party had a blood type found in 10% of the population. Suspect A does match this blood type, whereas the blood type of Suspect B is unknown.

(a) Given this new information, what is the probability that A is the guilty party?

(b) Given this new information, what is the probability that B's blood type matches that found at the crime scene?